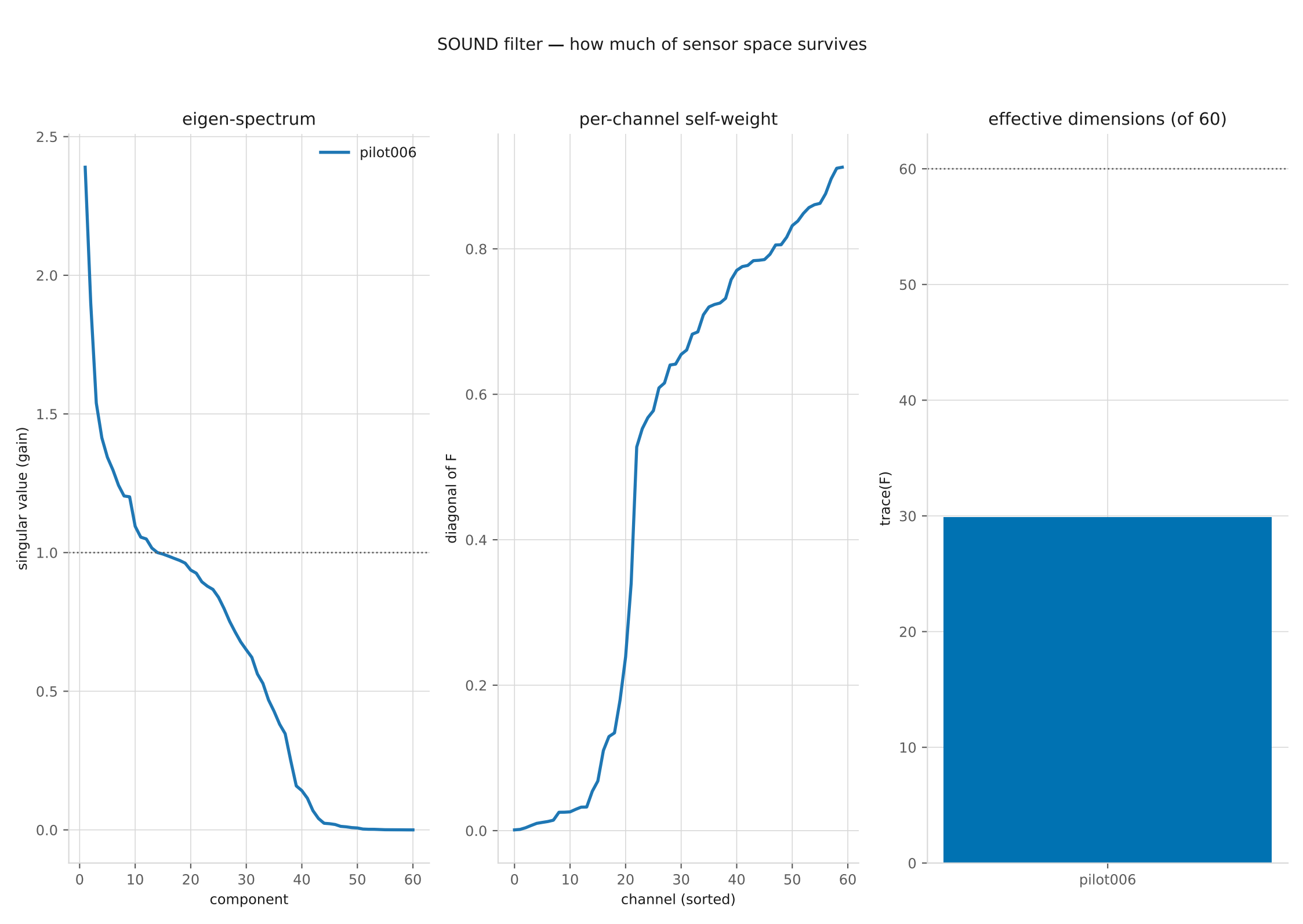
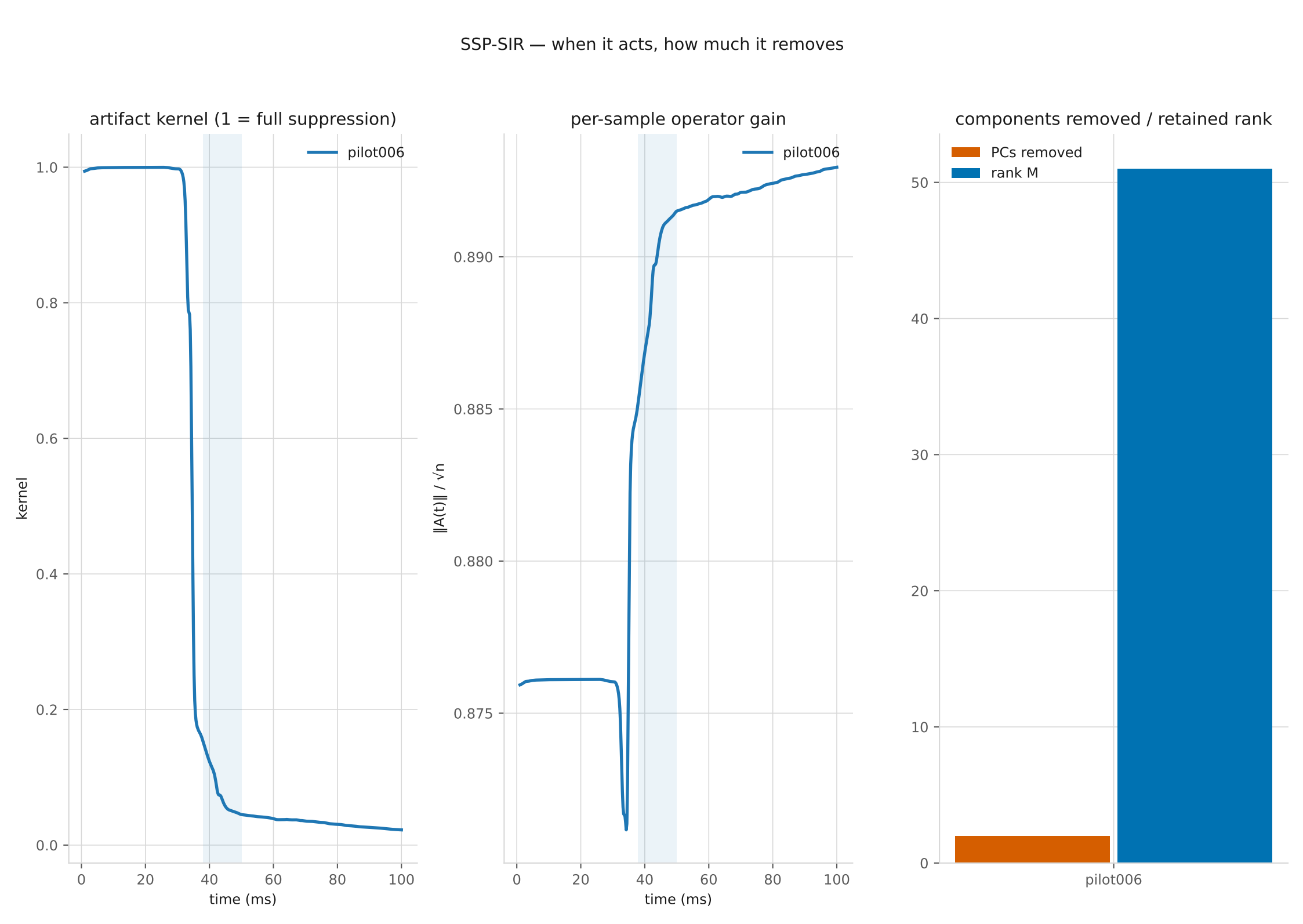
SOUND and SSP-SIR — online post-stimulus chain, replayed

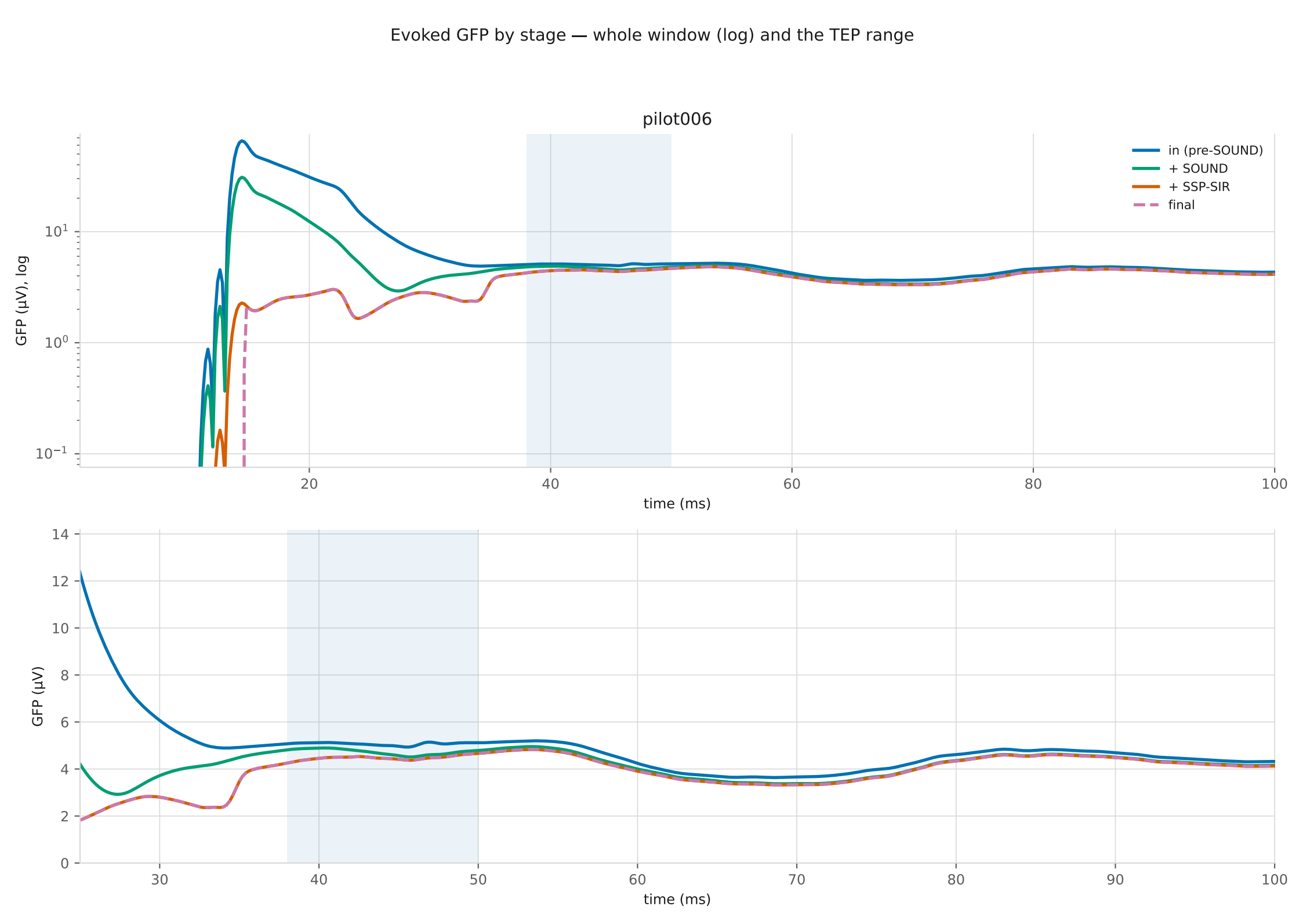
stages: pre_sound (SOUND input) -> sound -> sspsir -> final (== *_post_proc.npy)
trial groups: baseline, calibration, intervention, evaluation (intervention split into single / triplet; triplets only on the last page -- their +20 ms pu

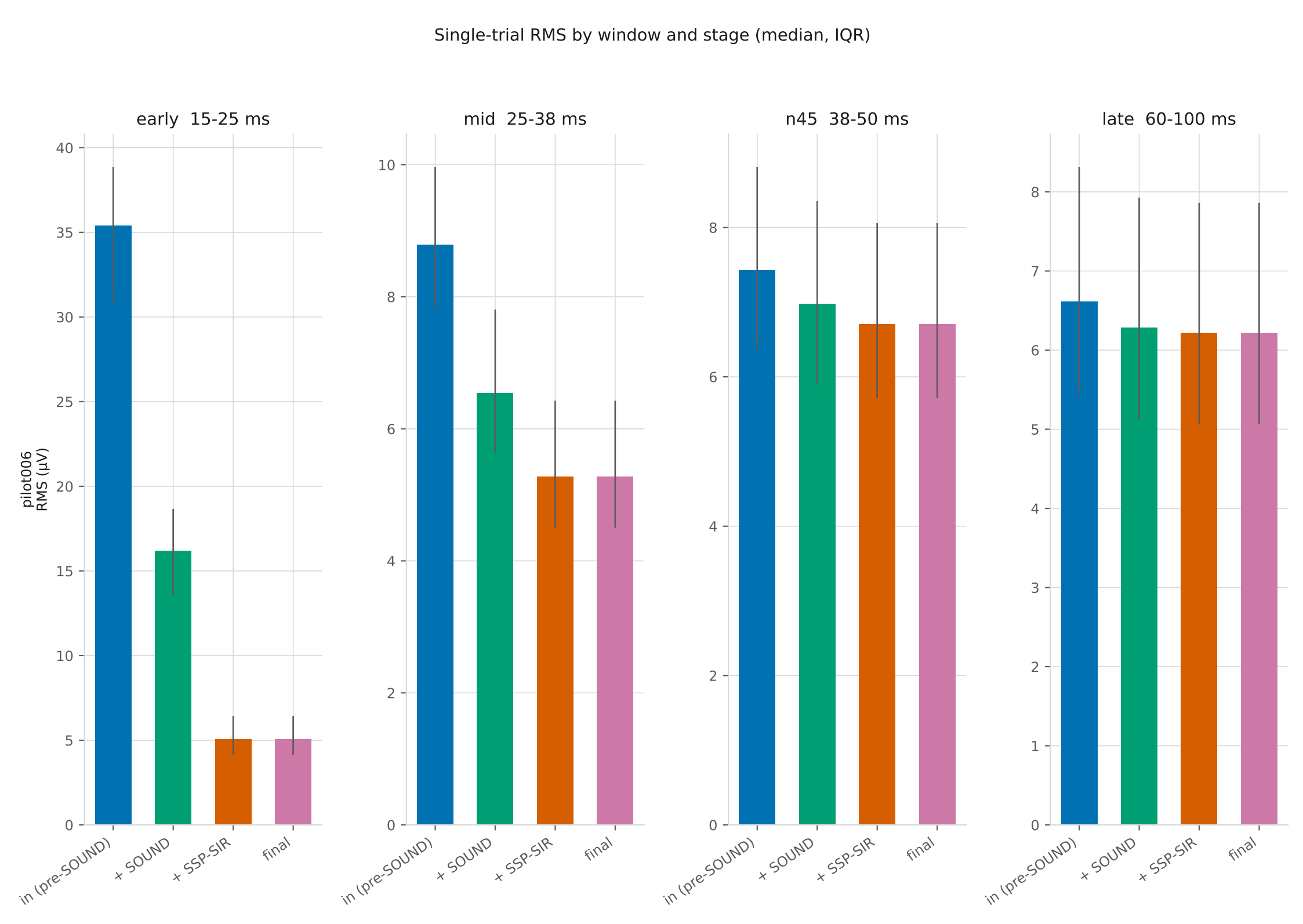
session	trials	replication	amp check	vs online	post Hz	crop win	fit win	SOUND dof	|F-I|	SSP PCs	rank M	kernel@fit	bad channels
pilot006	1425	6.37e-12	0.00e+00	1.11e-16	5000	38-50	42.4-48.0	29.9	0.829	2	51	0.080	C3,F3,FC1,FC3,FT7,T7,T

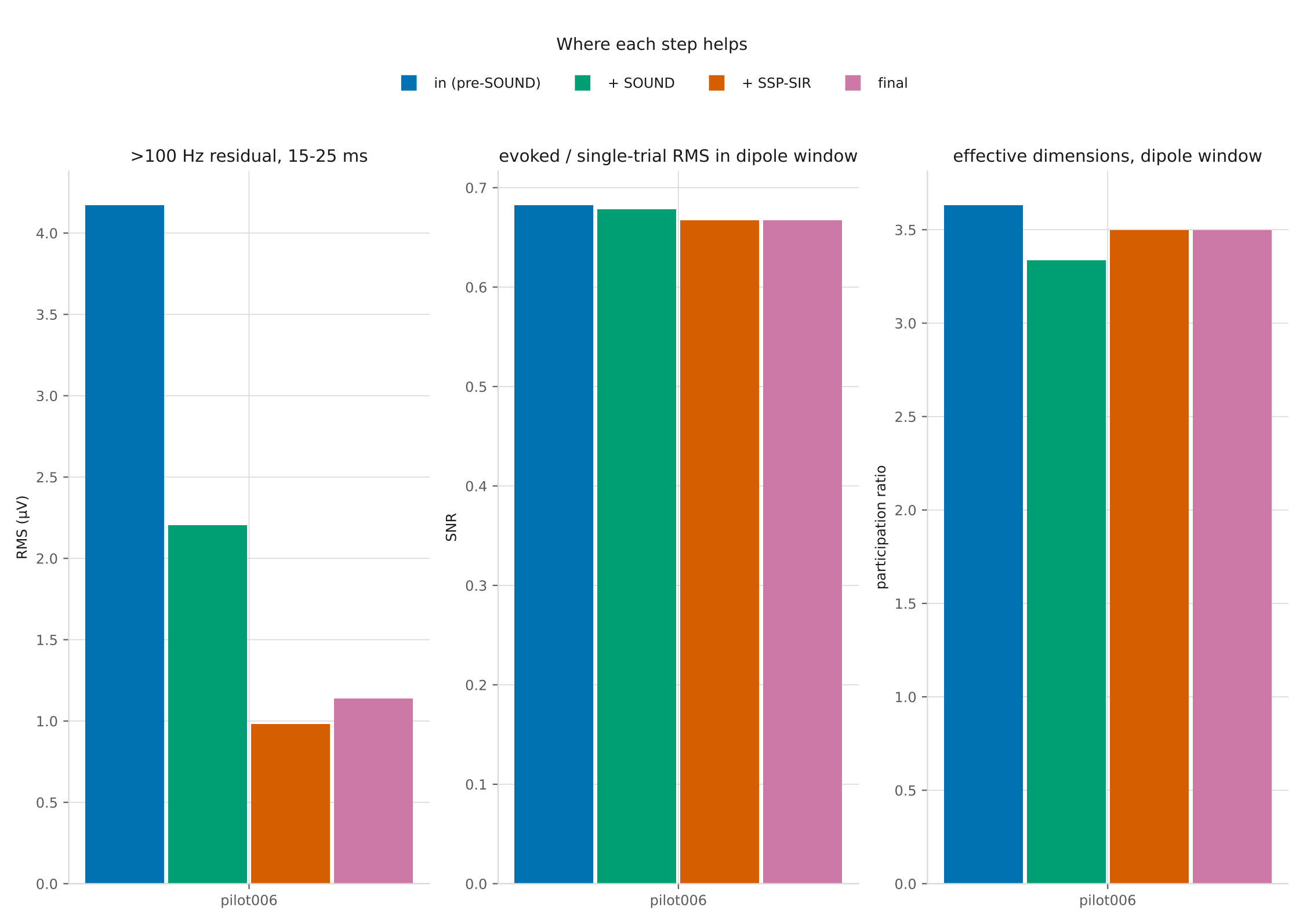
replication = max |replayed final − saved *_post_proc.npy| (expect ~1e-12 or 0)
amp check   = max |own dipole fit − DipoleFitter.fit_trial| (expect ~1e-15)
vs online   = max |fit_trial − tep_amplitude_raw in the trial table|
SOUND dof = trace of the filter, out of 60 channels; |F-I| = RMS distance from identity
post Hz = rate of the post-stimulus chain (from the calibration bundle); pilot006 500 Hz low-pass on

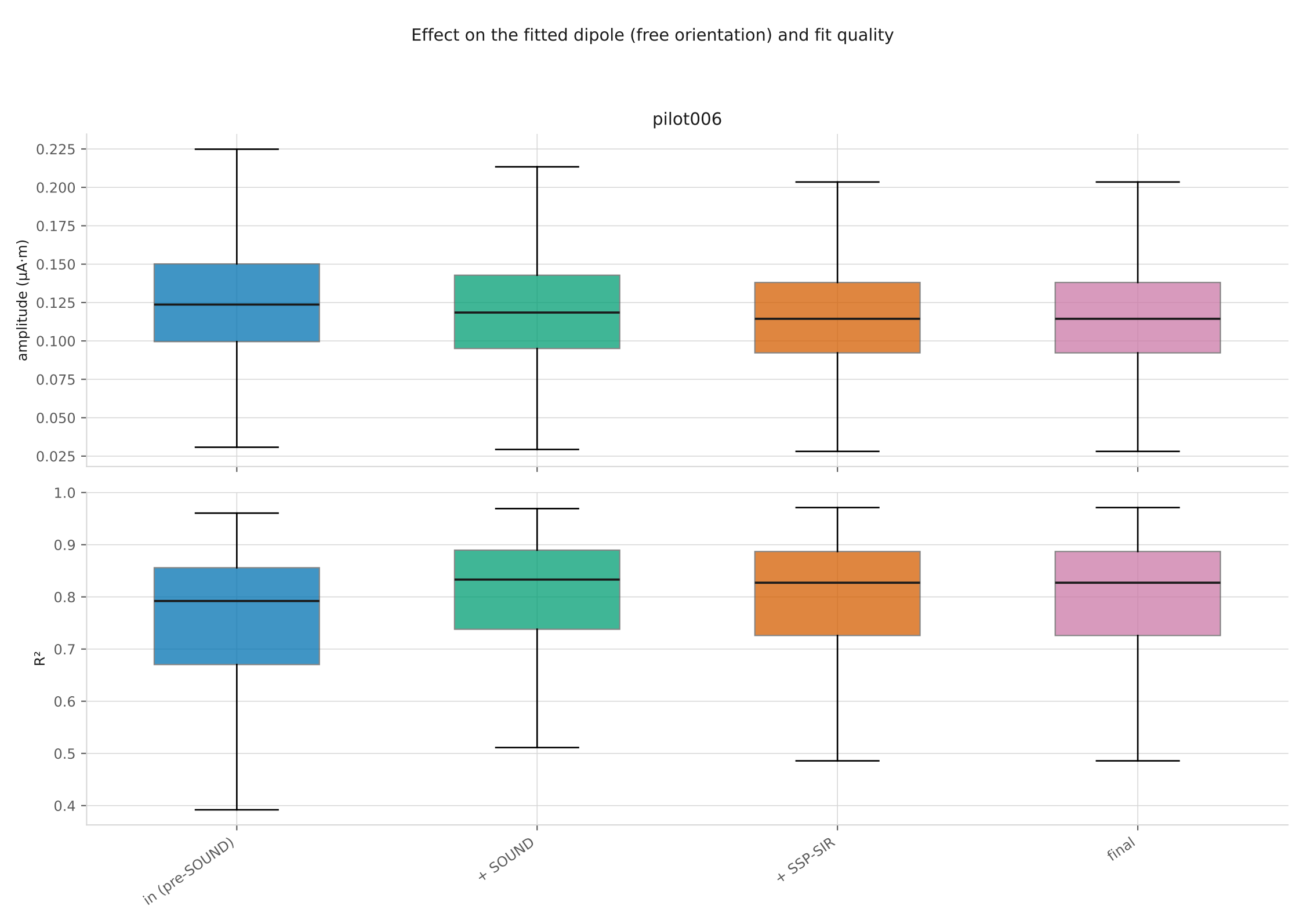

SOUND filter — how much of sensor space survives
eigen-spectrum
pilot006
singular value (gain)
component
per-channel self-weight
diagonal of F
channel (sorted)
effective dimensions (of 60)
trace(F)
pilot006

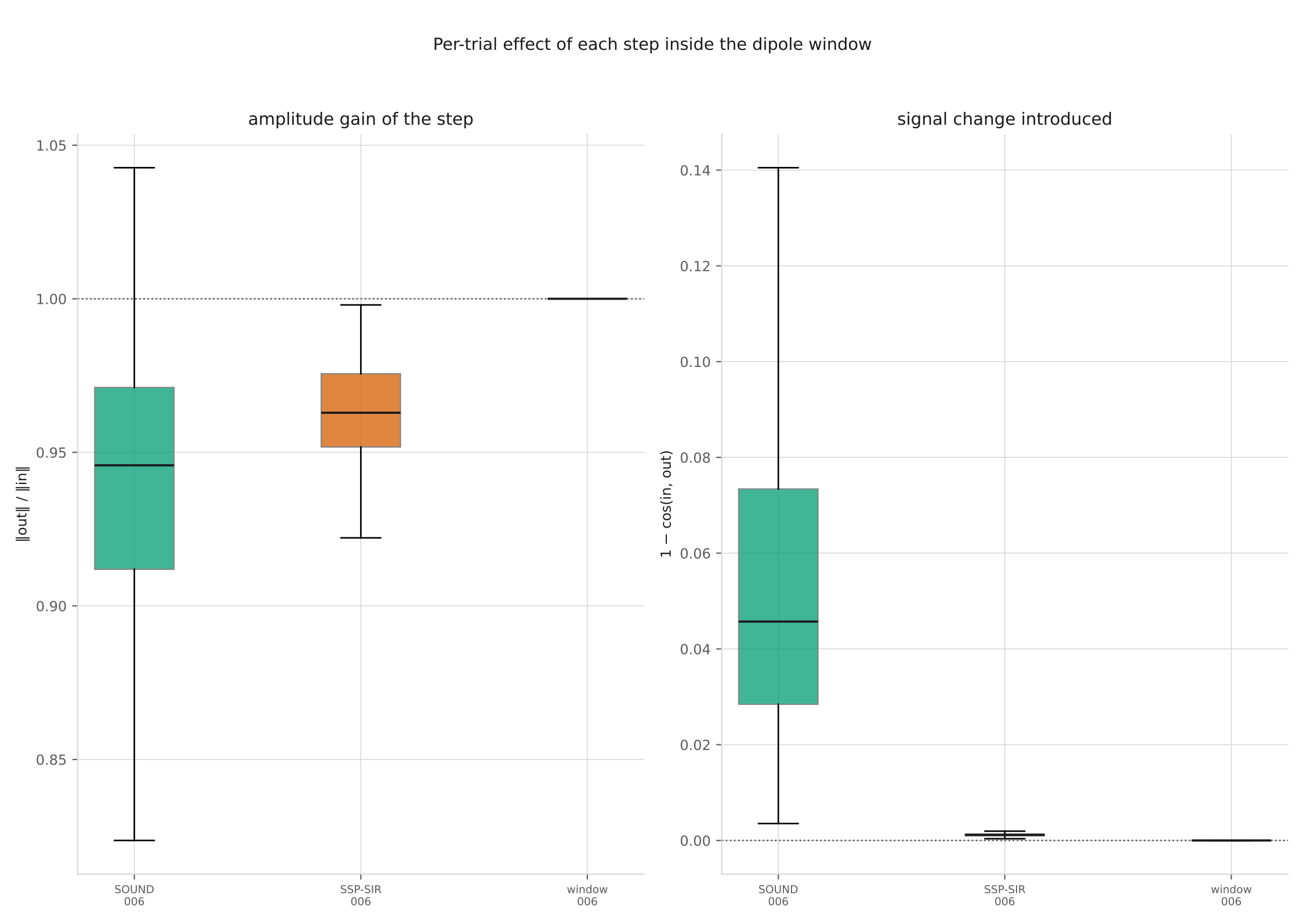

SSP-SIR — when it acts, how much it removes
artifact kernel (1 = full suppression)
pilot006
kernel
time (ms)
per-sample operator gain
pilot006
‖A(t)‖ / √n
time (ms)
components removed / retained rank
PCs removed
rank M
pilot006

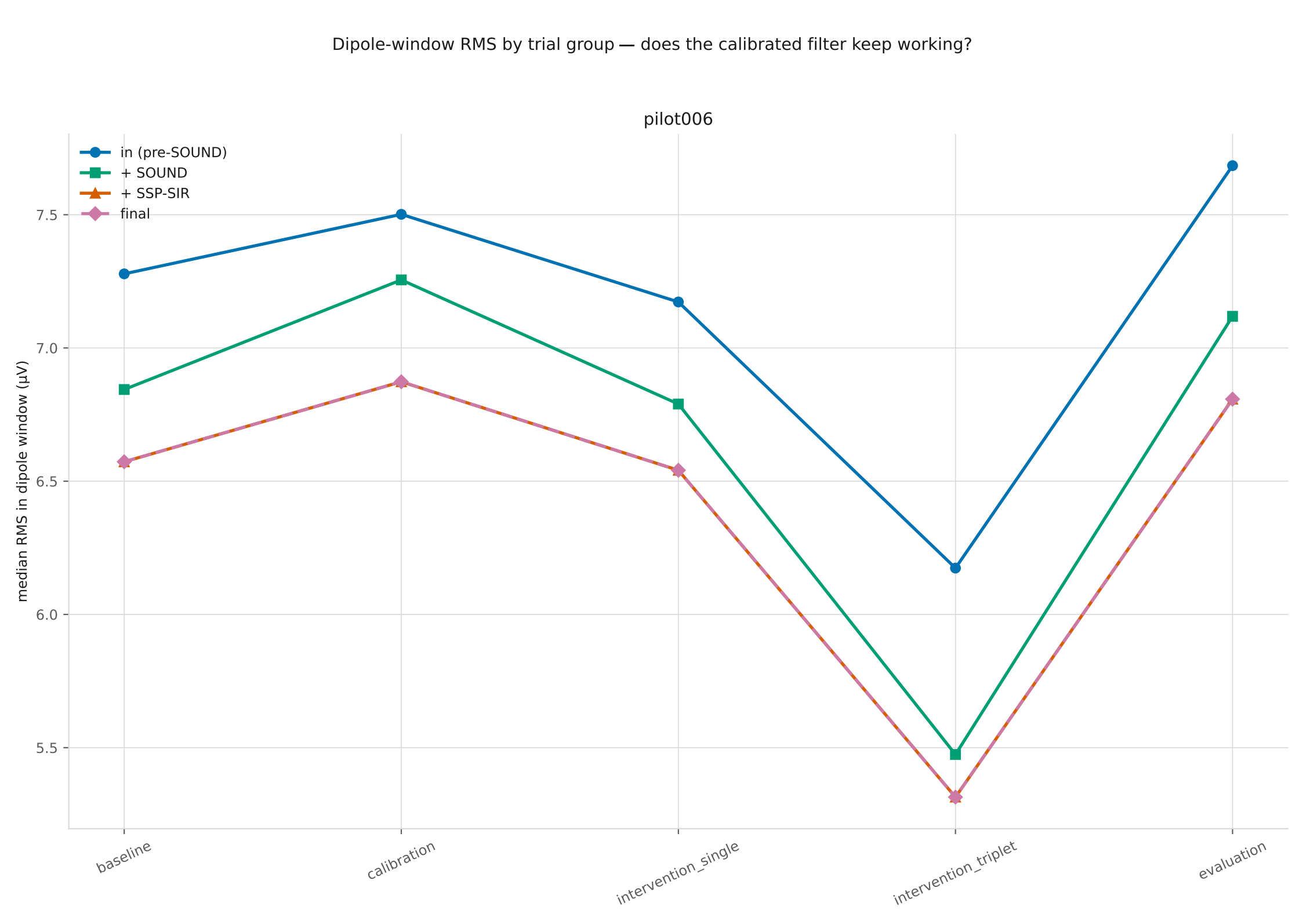

Evoked GFP by stage — whole window (log) and the TEP range
pilot006
in (pre-SOUND)
+ SOUND
+ SSP-SIR
final
GFP (µV), log
10^{1}
10^{0}
10^{-1}
20
40
60
80
100
time (ms)
GFP (µV)
14
12
10
8
6
4
2
0
30
40
50
60
70
80
90
100
time (ms)


Single-trial RMS by window and stage (median, IQR)
early 15-25 ms
mid 25-38 ms
n45 38-50 ms
late 60-100 ms
pilot006
RMS (µV)
in (pre-SOUND)
+ SOUND
+ SSP-SIR
final


Where each step helps
in (pre-SOUND)
+ SOUND
+ SSP-SIR
final
>100 Hz residual, 15-25 ms
RMS (µV)
4.0
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
pilot006
evoked / single-trial RMS in dipole window
SNR
0.7
0.6
0.5
0.4
0.3
0.2
0.1
0.0
pilot006
effective dimensions, dipole window
participation ratio
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
pilot006


Effect on the fitted dipole (free orientation) and fit quality
pilot006
amplitude (µA·m)
0.225
0.200
0.175
0.150
0.125
0.100
0.075
0.050
0.025
R²
1.0
0.9
0.8
0.7
0.6
0.5
0.4
in (pre-SOUND)
+ SOUND
+ SSP-SIR
final


Per-trial effect of each step inside the dipole window
amplitude gain of the step
‖out‖ / ‖in‖
1.05
1.00
0.95
0.90
0.85
SOUND
006
SSP-SIR
006
window
006
signal change introduced
1 − cos(in, out)
0.14
0.12
0.10
0.08
0.06
0.04
0.02
0.00
SOUND
006
SSP-SIR
006
window
006


Dipole-window RMS by trial group — does the calibrated filter keep working?
pilot006
in (pre-SOUND)
+ SOUND
+ SSP-SIR
final
median RMS in dipole window (µV)
7.5
7.0
6.5
6.0
5.5
baseline
calibration
intervention_single
intervention_triplet
evaluation